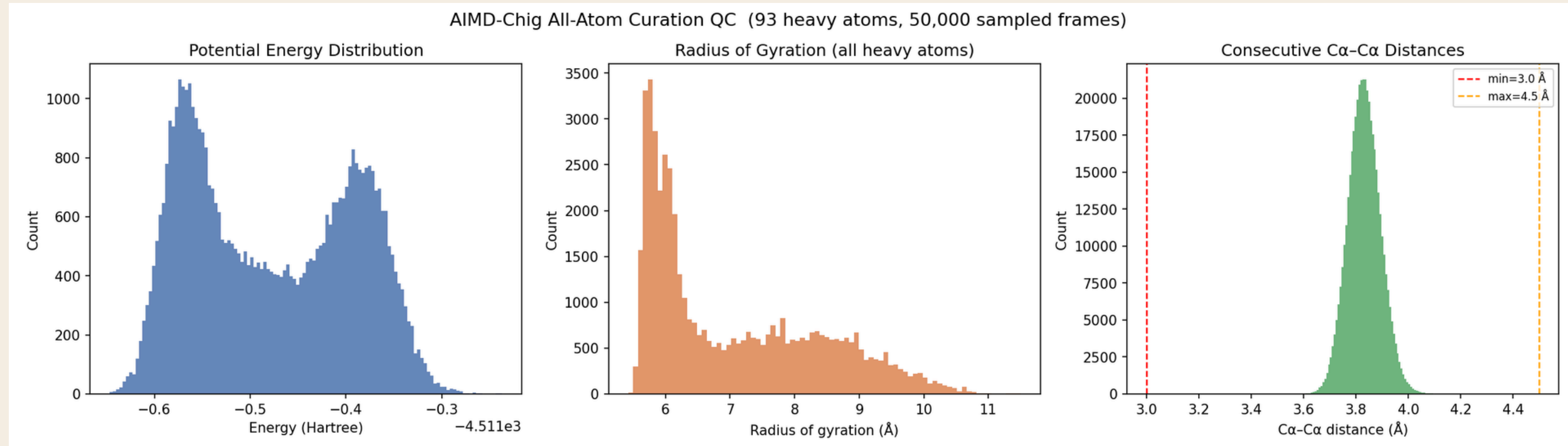


Part 2

Implementing custom diffusion models

Dataset

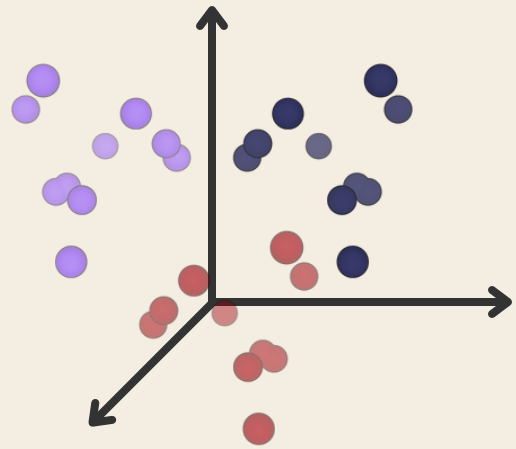


- Ab initio MD dataset of Chignolin (Liu *et al.*, 2023, Scientific Data)
- 100K structures including energy labels

Model Architectures

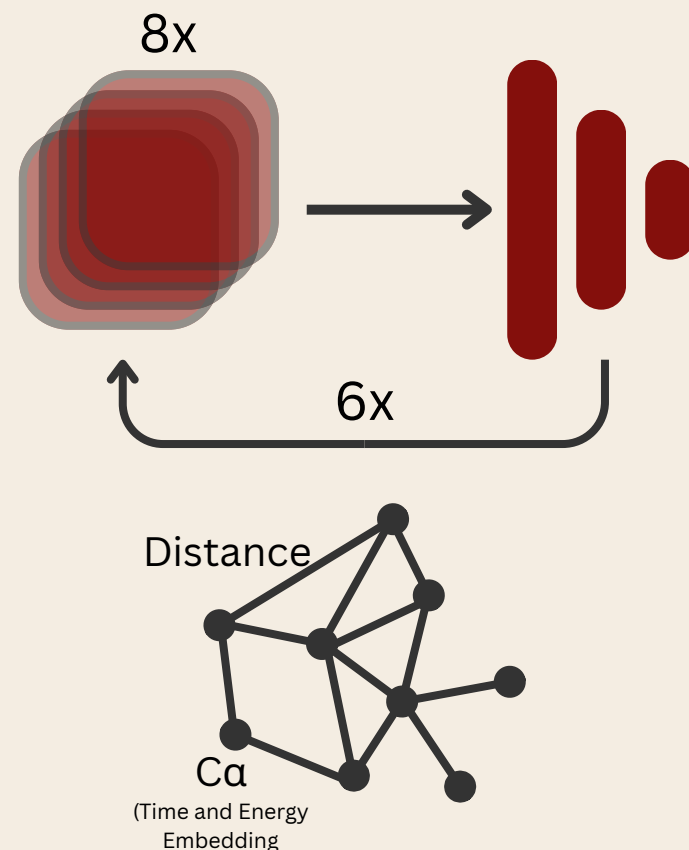
Data set

- **Native Dataset**
- **Dataset Augmentation**
 - Translation
 - Rotation



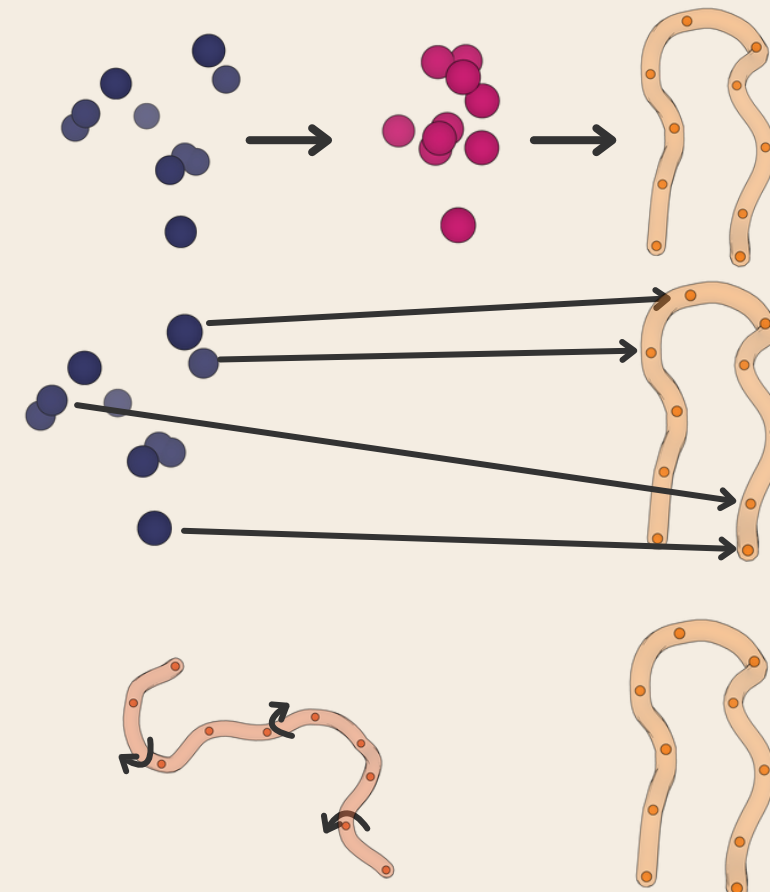
Model Architecture

- **Transformer**
 - 8 Heads
 - 6 Layers
 - 7.45 M params
- **EGNN**
 - 0.44 M params
 - 1.7 M params



Sampler

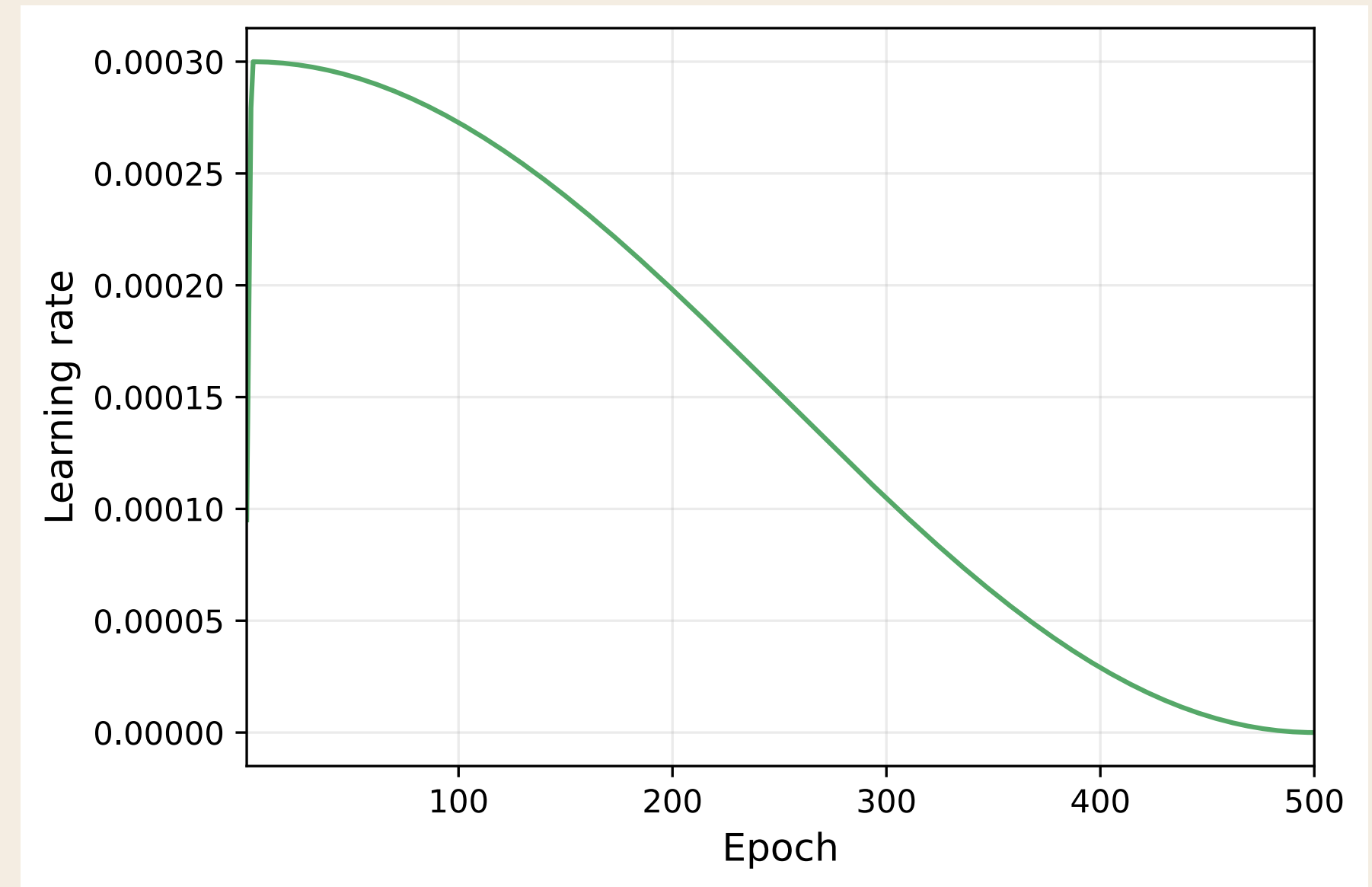
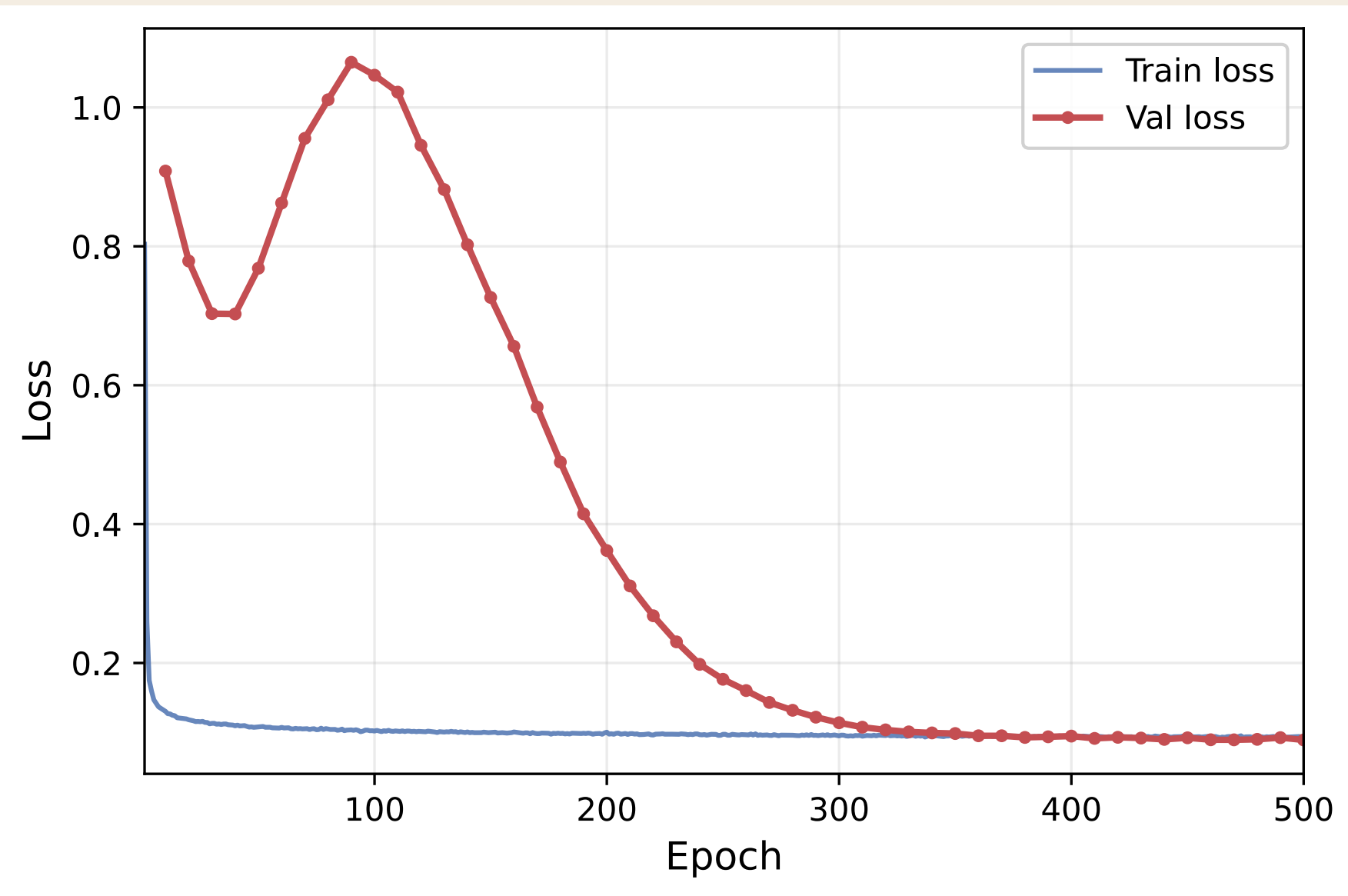
- **DDPM**
 - denoising diffusion probabilistic models
- **ZeroCoM**
 - zero center of mass flow matching
- **Torsion flow matching**



Constraints

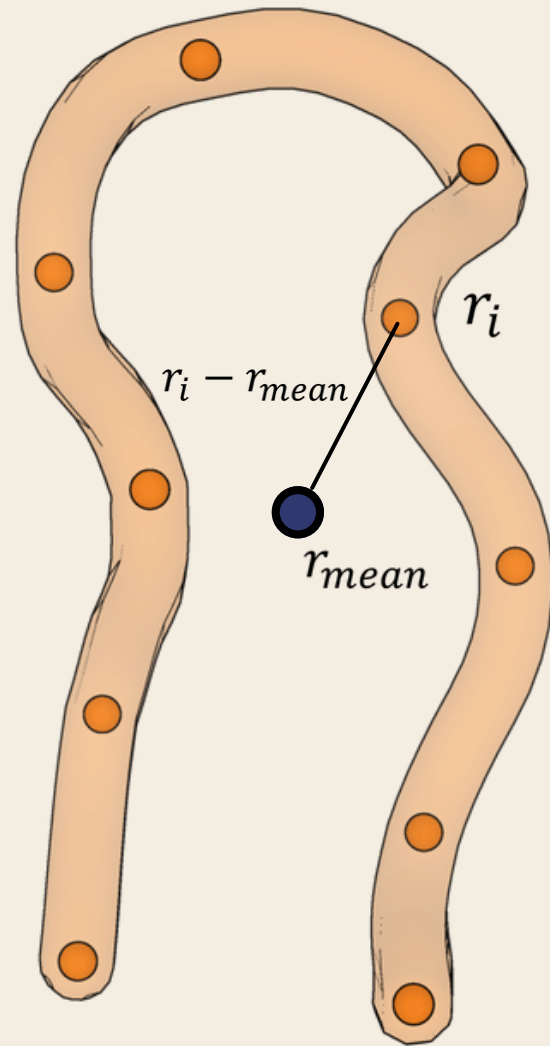
- **Physical constraints**
- **Energy conditioning**

Model Training



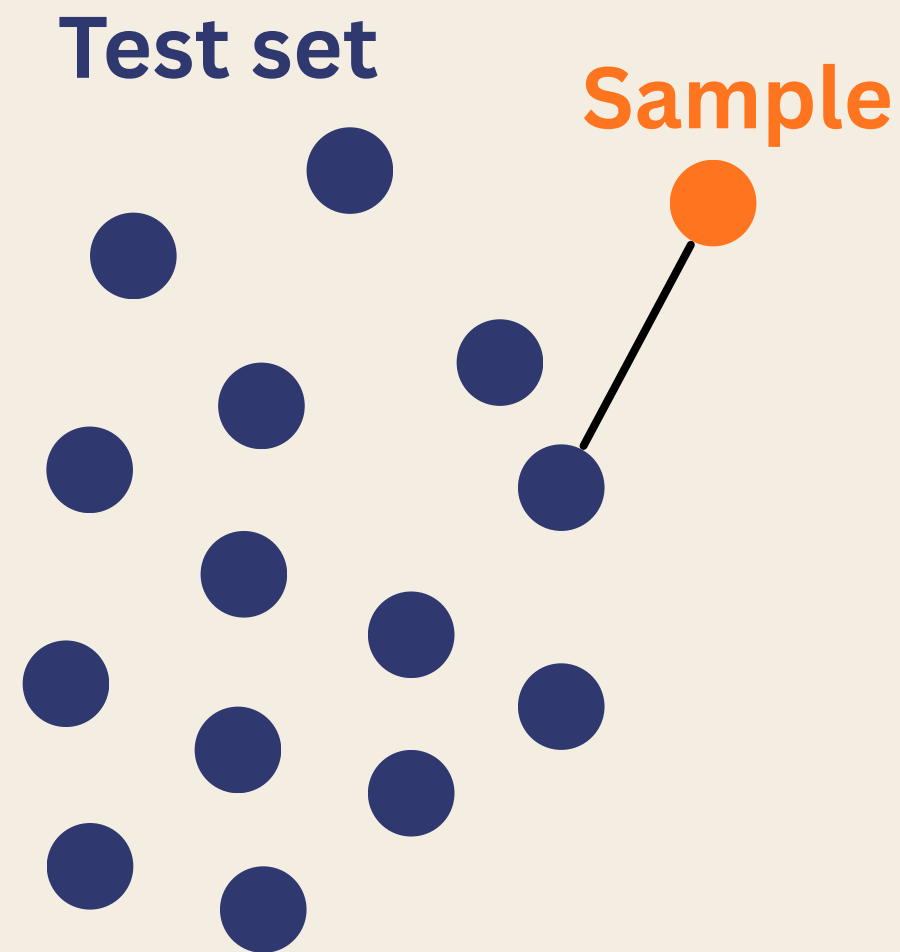
Evaluation Metrics

Radius of gyration

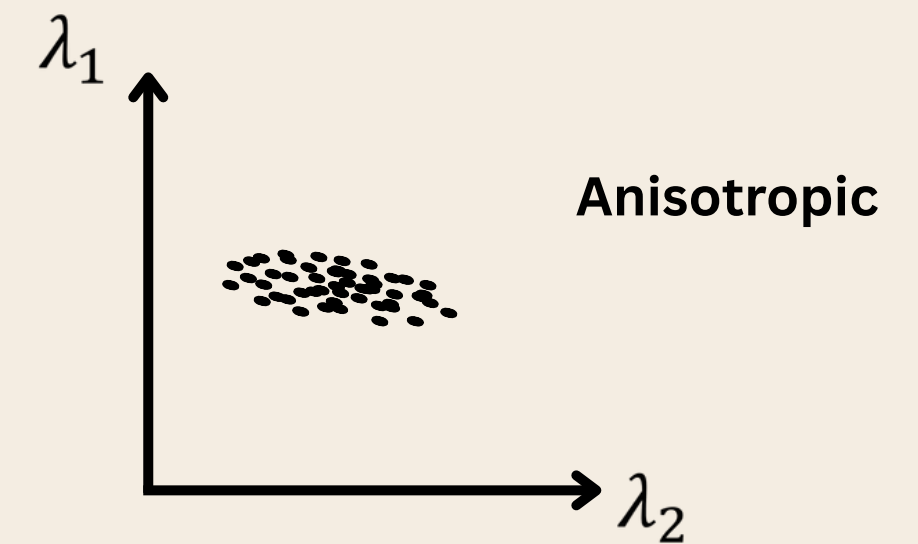
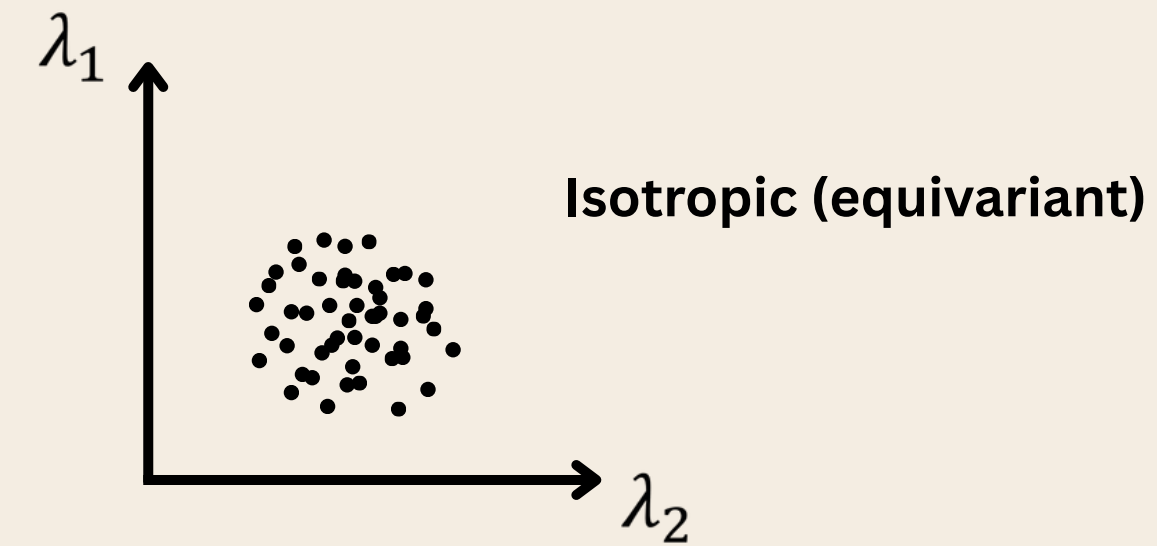


$$R_g = \sqrt{\frac{1}{N} \sum_{i=0}^N \|r_i - r_{mean}\|^2}$$

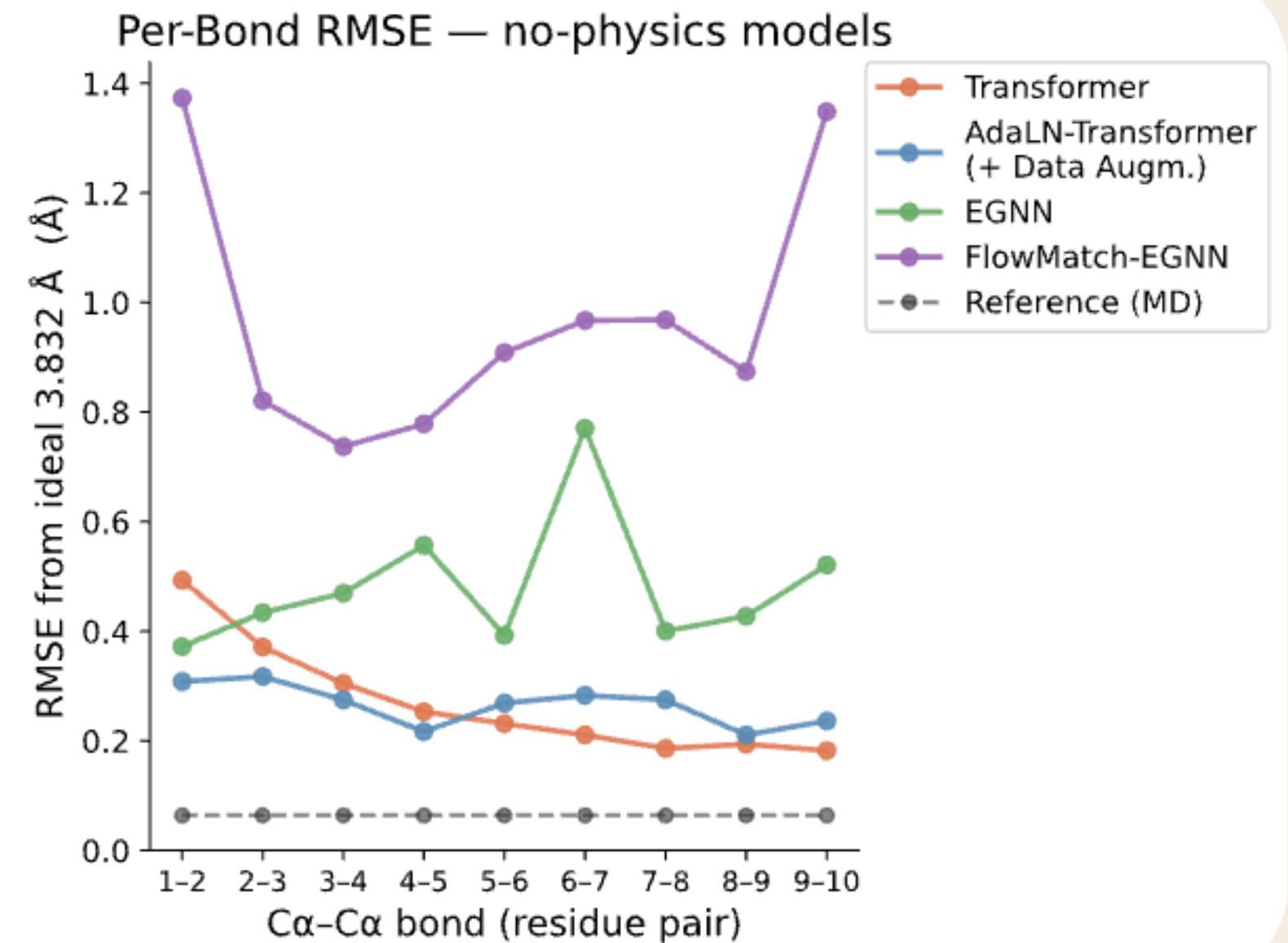
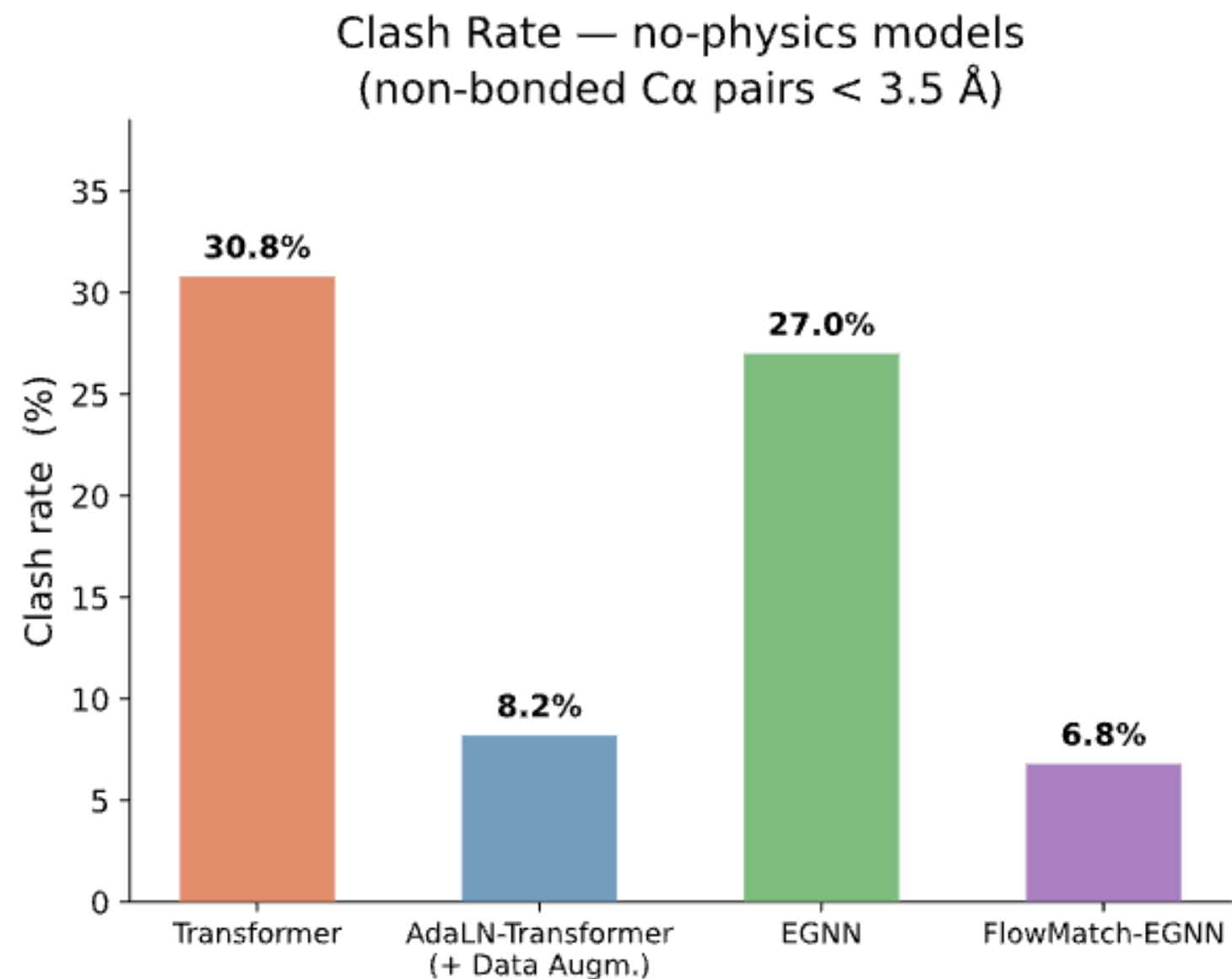
Nearest Neighbor Distance (Novelty)



Isotropy (Equivariance)

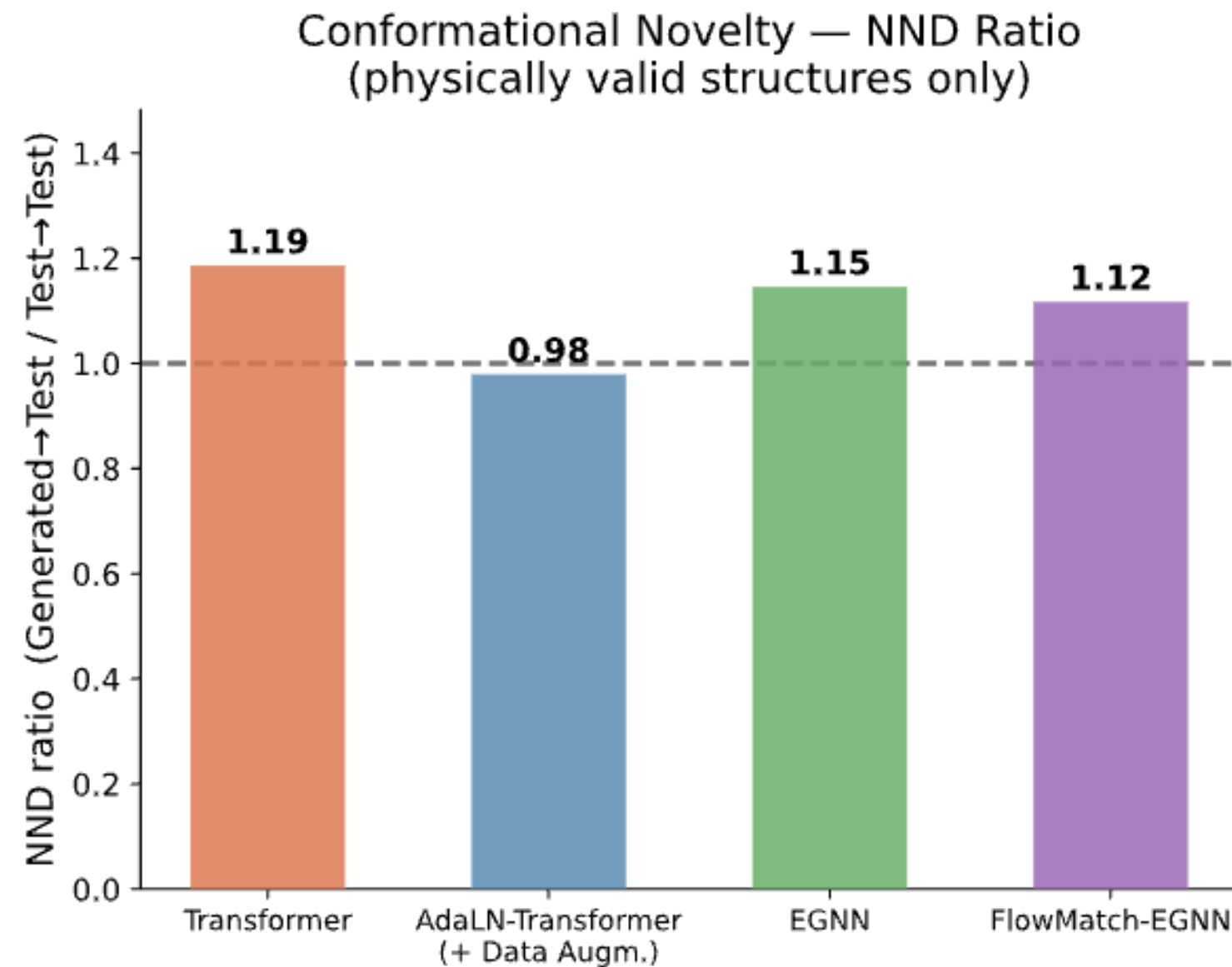


Physical Validity of C α Designs

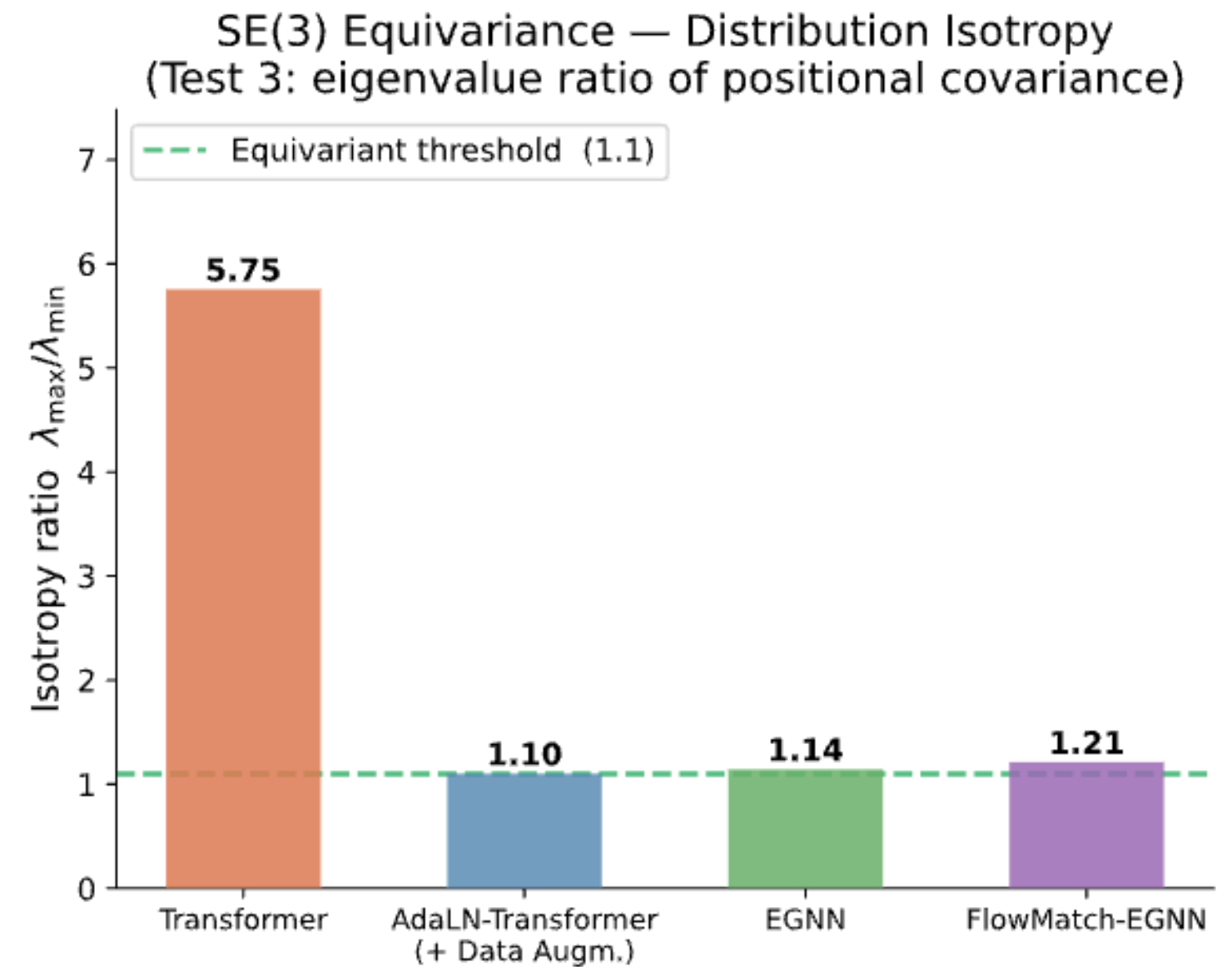


→ Models without physical constraints show low design validity

Novelty and SE(3)Equivariance



→ Designed structures show some novelty



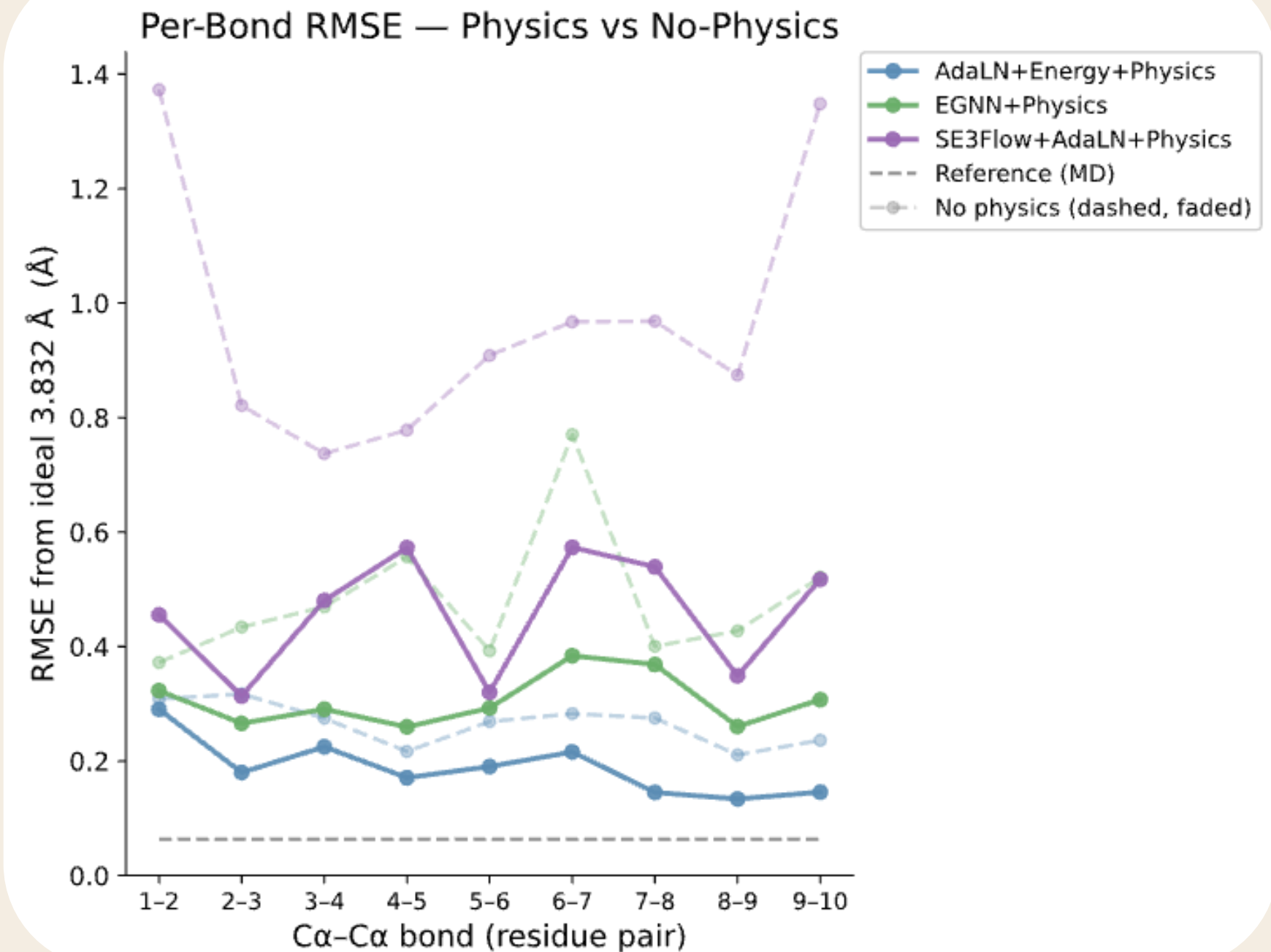
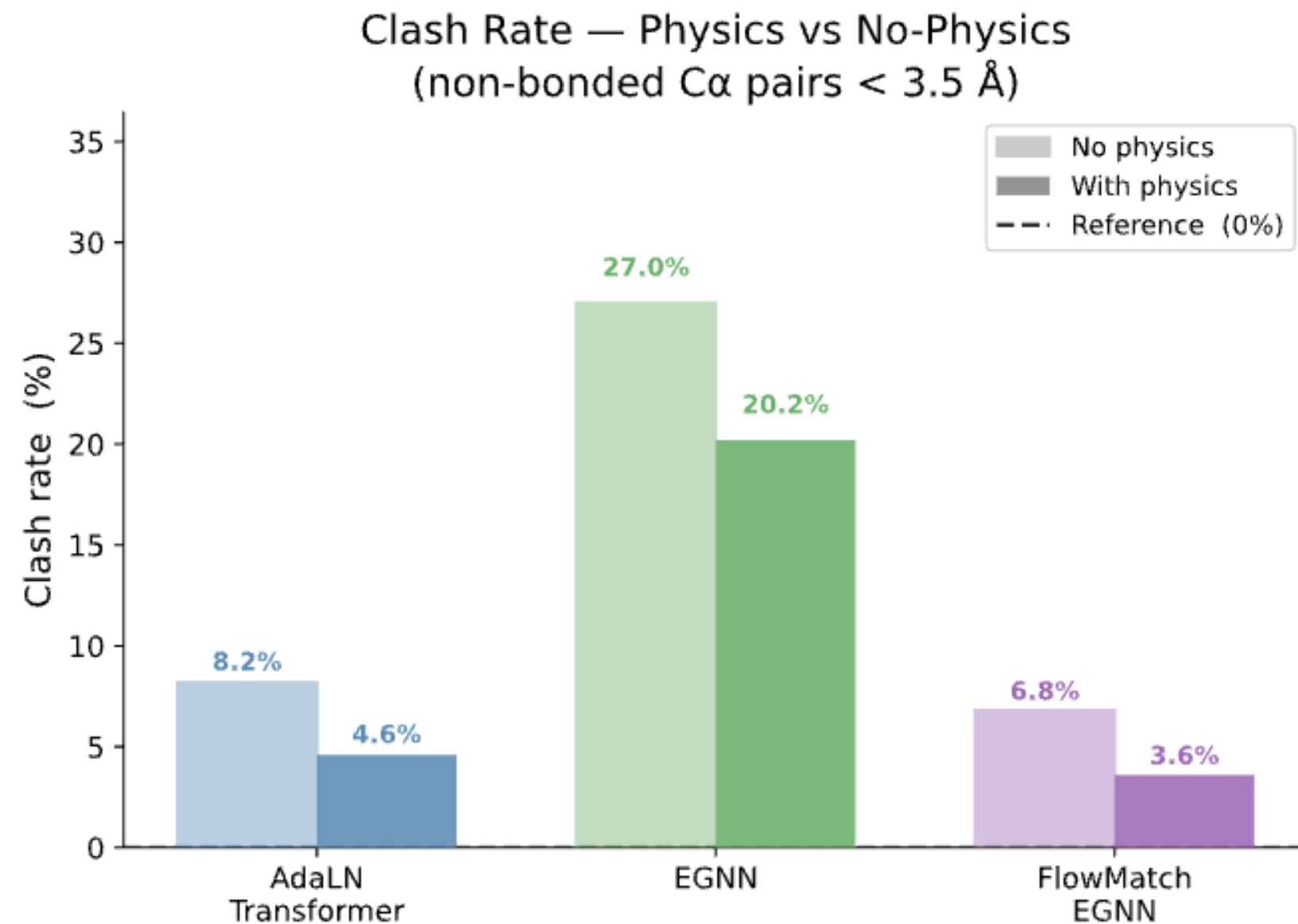
→ Data Augmentation and EGNNs enable SE(3)-equivariance

Physical Constraints Loss

1. **Bond Length:** MSE between bond length and ideal length between C α 's (3.83 Å)
2. **Steric Clash:** Quadratic repulsion between non-bonded C α below 3.5Å distance
3. **Bond Angle:** Penalizes deviation from realistic angles between three consecutive C α 's

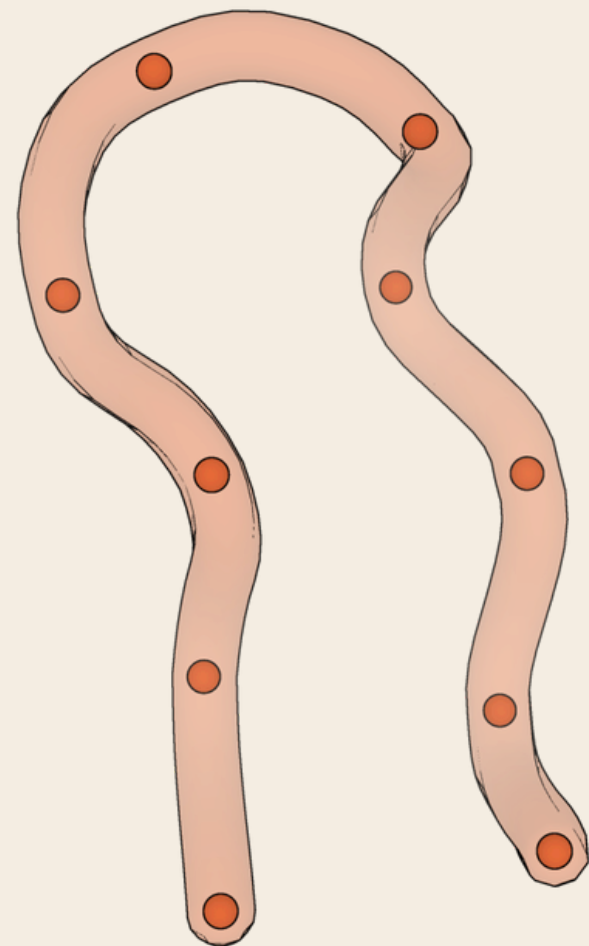
$$\text{Total Loss} = \text{MSE}(\text{predicted noise, true noise}) + t * 0.05 * \text{physics_loss}(\text{predicted structure})$$

Physics-Constrained Models



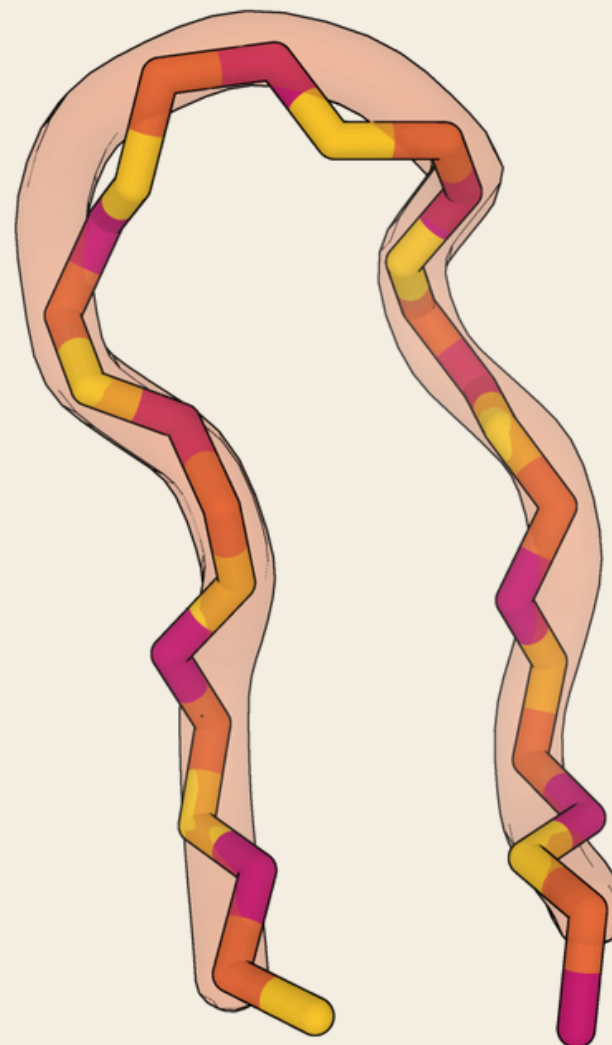
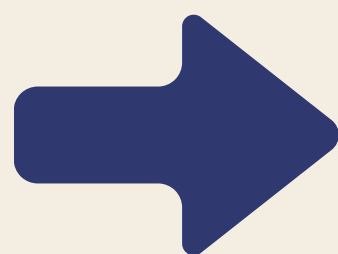
→ Physically constrained models design more plausible structures

Beyond Ca models



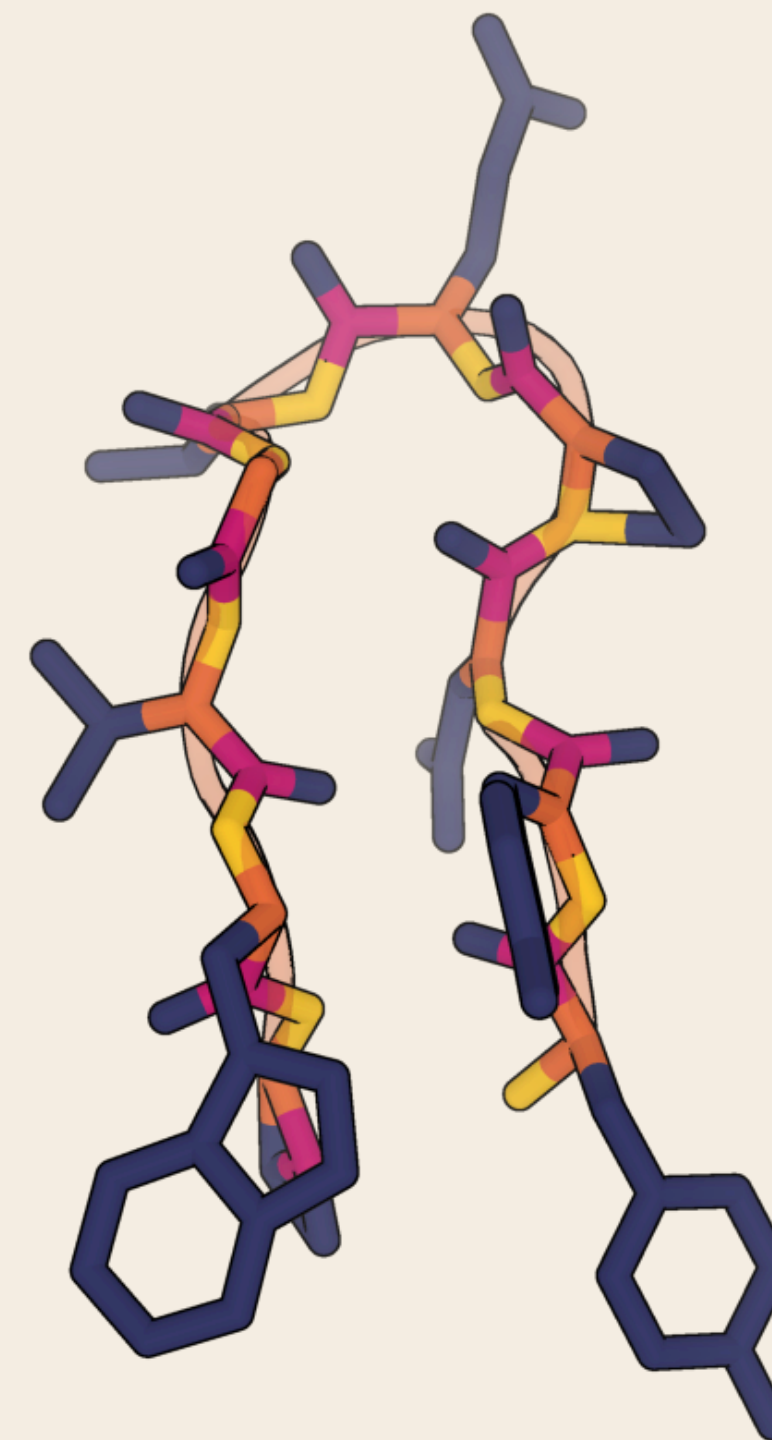
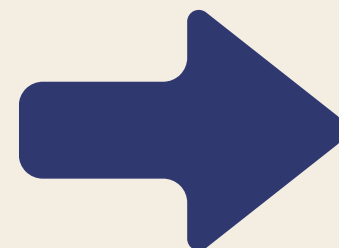
Ca only

Previous Models



Backbone

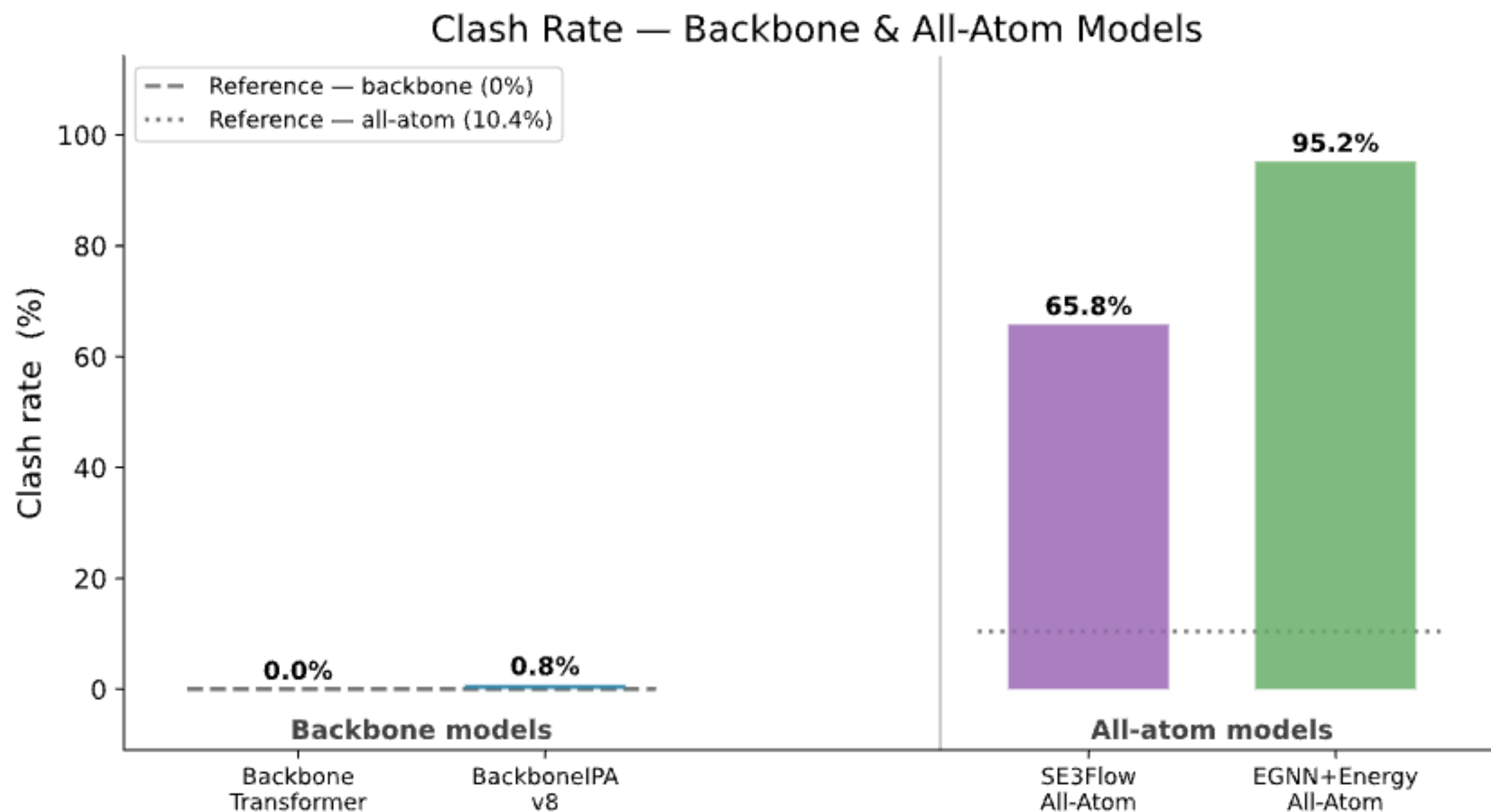
Transformer
TorsionTransformer



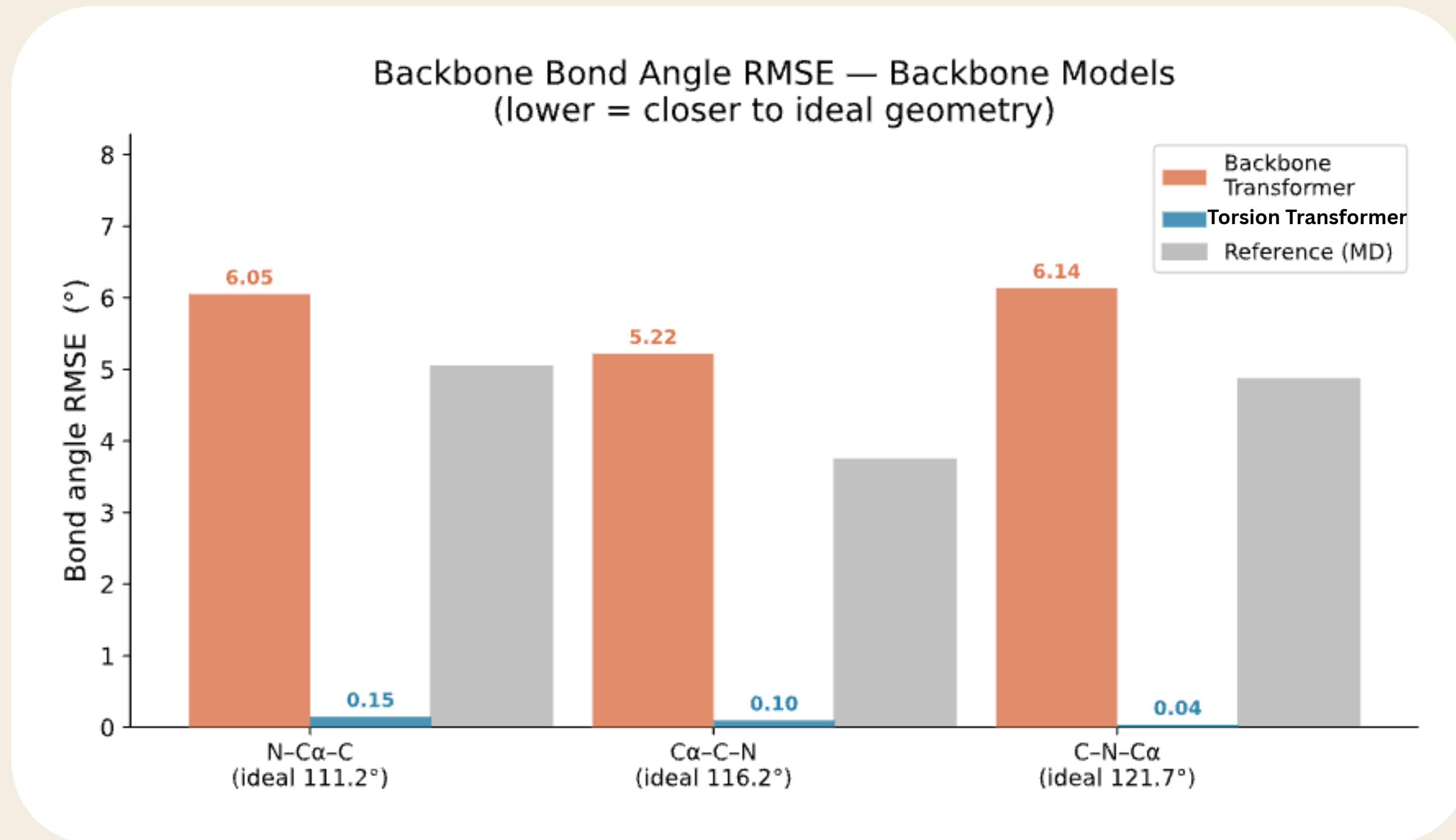
All-Atom

EGNN
FlowMatchTransformer

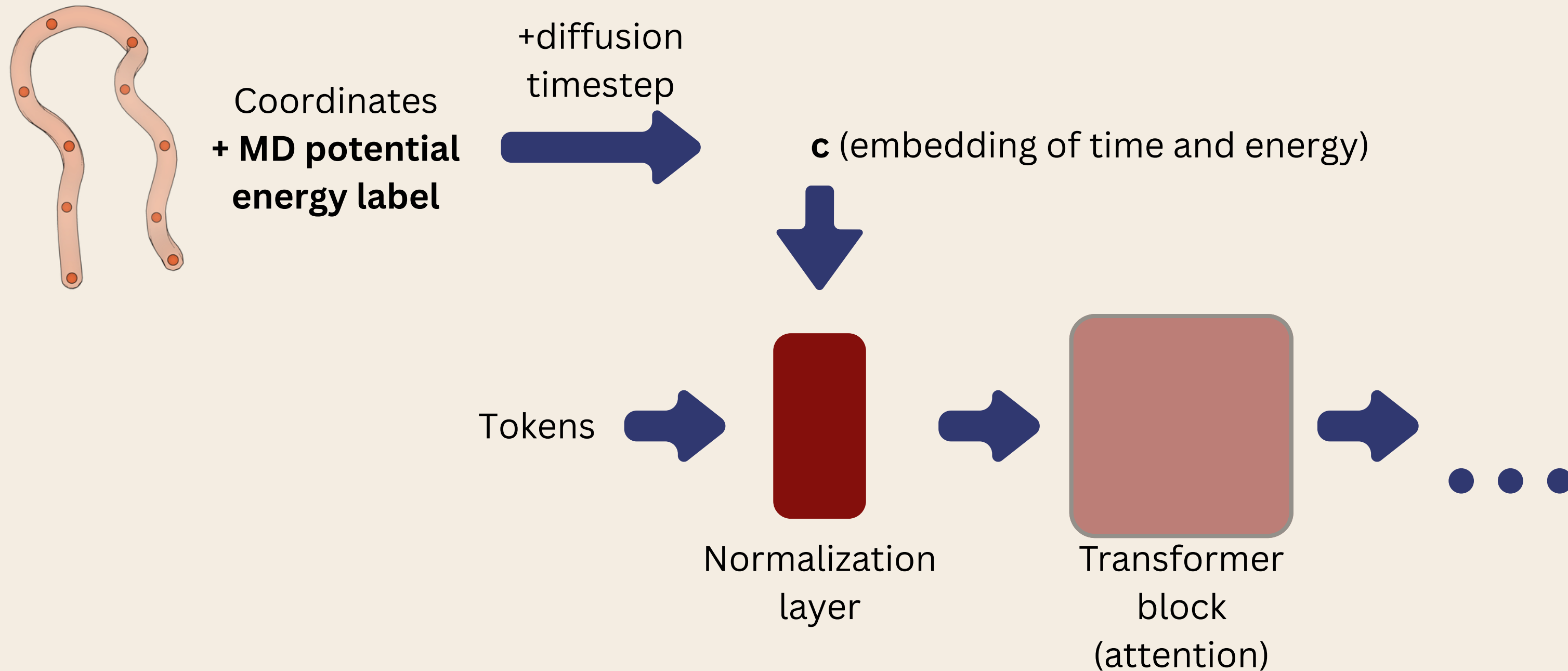
Beyond Ca models



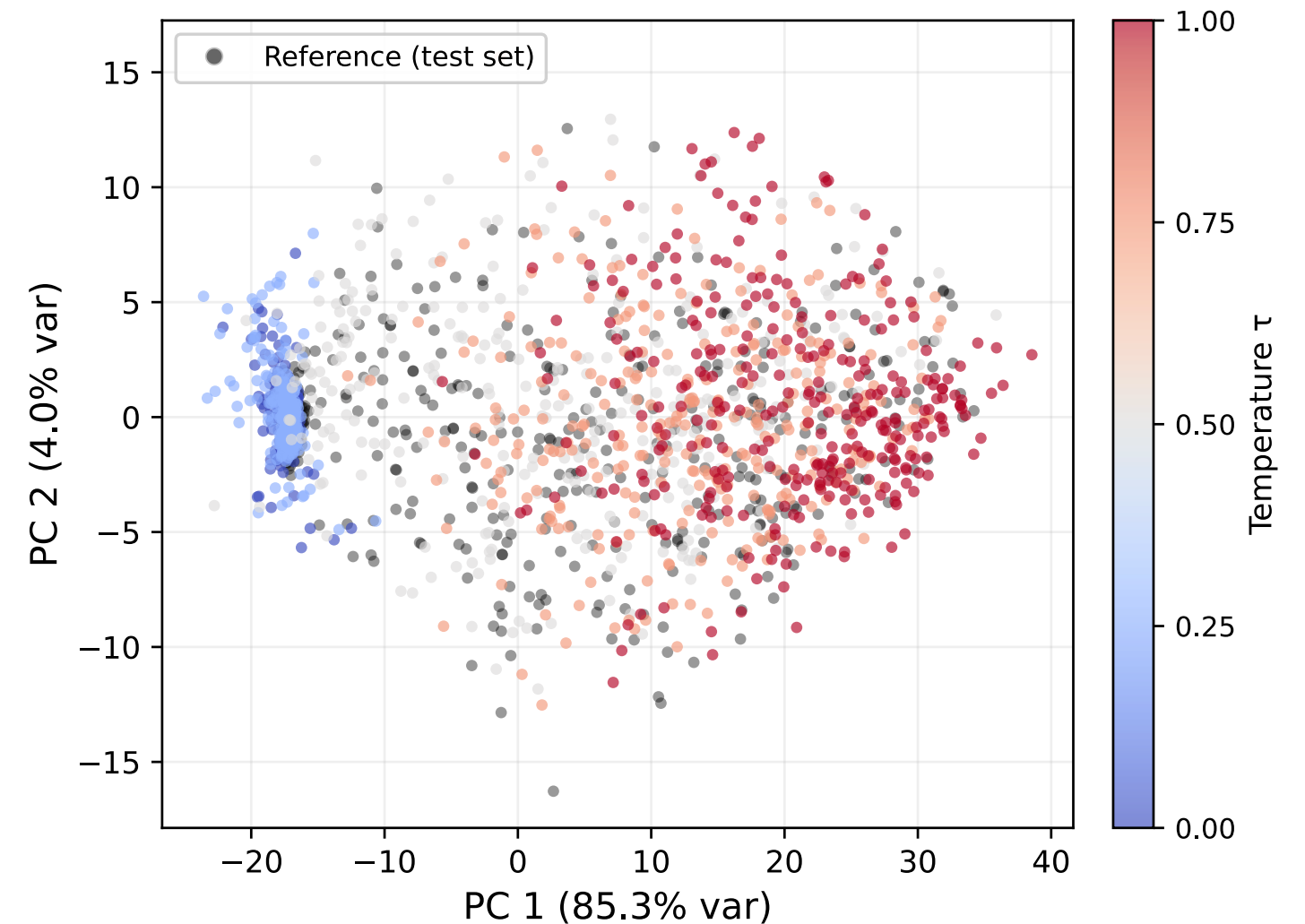
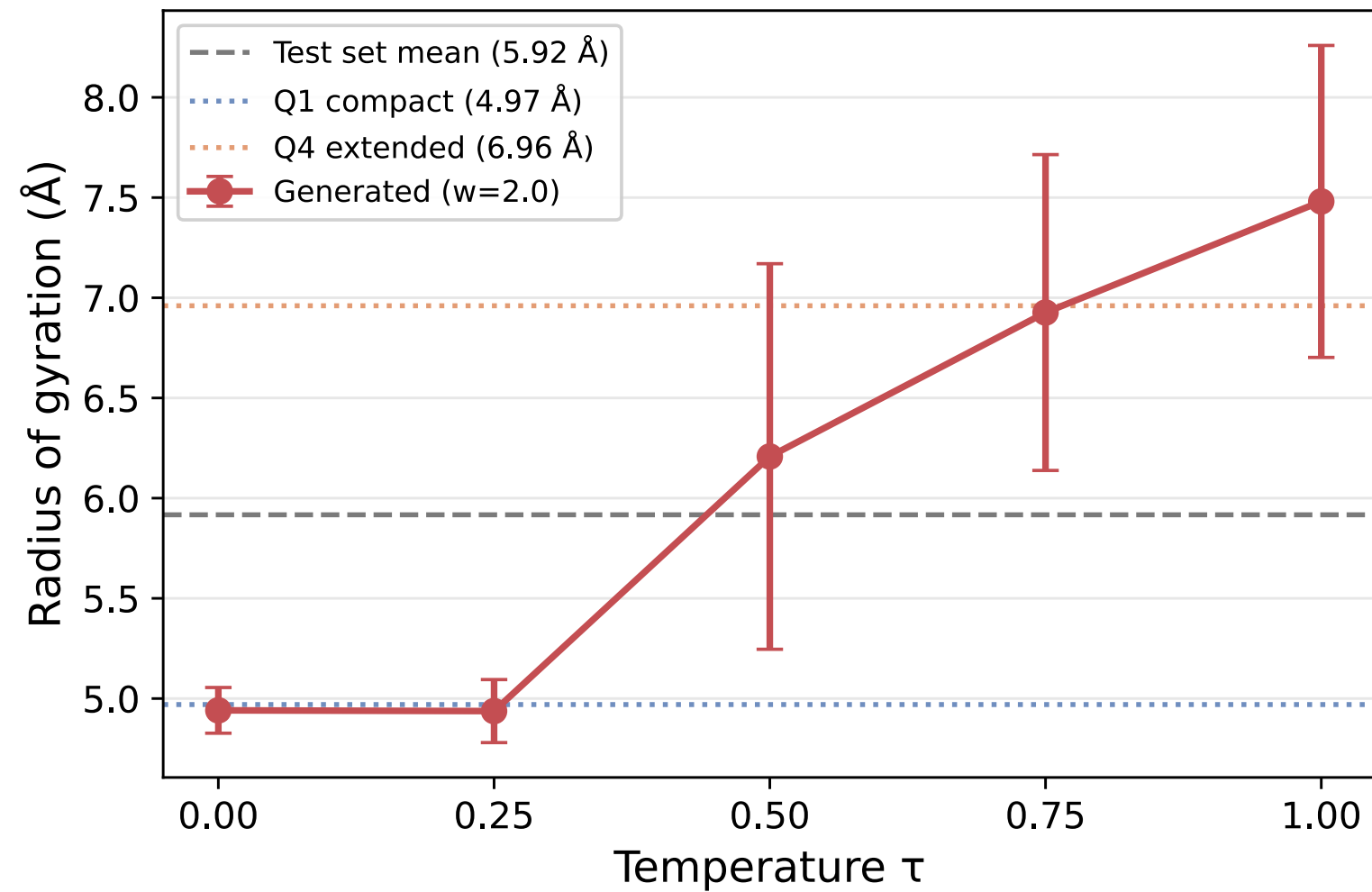
Beyond Ca models



Energy Conditioning



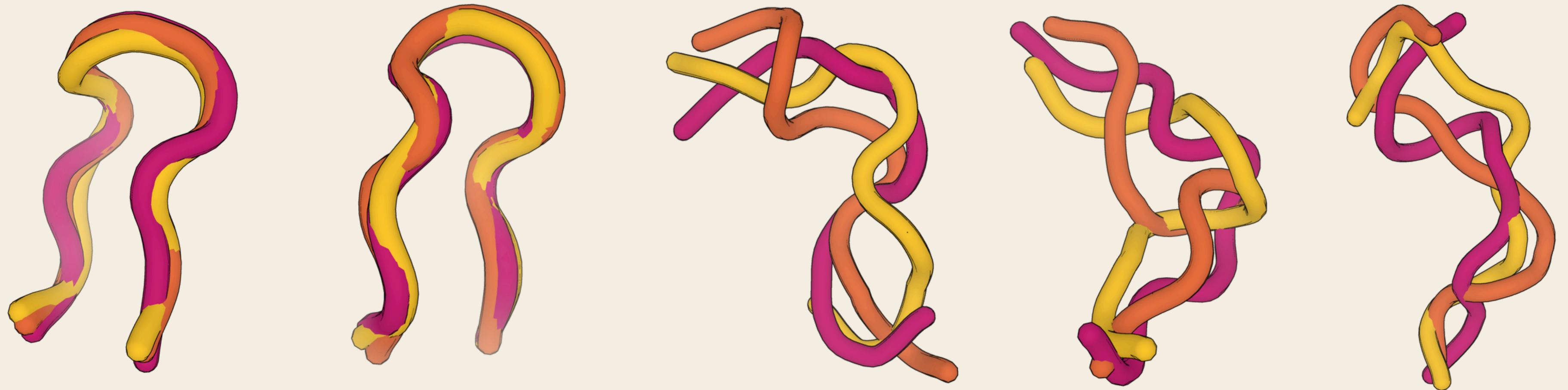
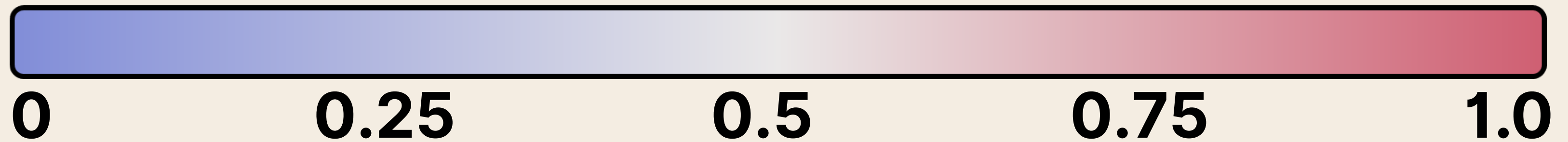
Testing Energy Conditioning



→ High energy samples cause shift in Rg and increase structural diversity

Testing Energy Conditioning

Temperature τ



3 representative structures out of 5 generated samples

Bottom-line

- Coordinate representation is the dominant factor for bond validity
- Physics loss yields best improvements
- Higher number of parameters risks memorization
- All-atom approaches require more sophisticated models
- All models over-sample conformational space compared to test set

Thank you for your

Attention Is All You Need
